

RPent tool server

LLM planner

coding agent, e.g. Codex or Claude Code
context window

instruction ℓ + live observations

text memory M : task recipes, lessons

L_0 keyframes, before planning 

L_1 phase map, L_2 moments (text)

L_3 clip of node n , on demand 

L_3 clip of node n' , on demand 

⋮

one episode



query node n

text / frames

tool call

result, images

demonstration tools

view_task_video $\rightarrow L_0$

view_task_map $\rightarrow L_1, L_2$

view_demo_clip(n , cam) $\rightarrow L_3$

read-only; viewing budget B per node

read

robot tools

VLA: $\pi_{0.5}$ (pi0_pick)

analytic: move_to, release, ...

perception: SAM 3, back_project

actions

obs.

demonstration D of the task
frames, robot and object states

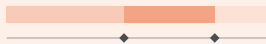
offline: segment at sub-events

demonstration hierarchy

one per base task



L_0 event-driven keyframes



L_1 phases

L_2 moments



L_3 clips (any phase/moment, camera)

environment



LIBERO-PRO, LIBERO-Plus...

layout, goal, camera,
lighting or noise differ
from the demonstration